

Mikiri: Knowing When a Frozen Go Engine Has Searched Enough

Anonymous submission

Abstract

A Go engine in the AlphaZero family gives every move the same search budget. We study how much stronger a frozen engine becomes when only that allocation changes. Mikiri wraps KataGo, leaves its network and search untouched, and climbs a short ladder of visit budgets, stopping at the first rung where the winrate still at stake no longer pays for the next rung. We build a dataset of 6,000 positions searched at seven budgets, with every candidate move scored by an independent forced search, and use it to train the stopping rule, to forecast match strength, and to compare against ten published stopping rules re-implemented on the same engine. Offline, Mikiri matches uniform search at 1.8 to 2.1 times the visits, ahead of the best prior rule by 0.54 [0.28, 0.84] in a paired test. In play against KataGo at 200 visits per move with the same network, it gains +228 Elo (95% interval +203 to +254) at equal visits over 1,000 games, and it draws level with KataGo at twice its visits, where doubling KataGo’s own budget gains +237 and the strongest prior rule, played the same way, gains +135. Dividing the predicted stake by the price of the next rung is the decisive ingredient: it turns stopping signals that are worse than no gate into gains, and it improves prior learned rules as well. We also report two negative results: approximate retrieval of stored searches does not transfer between similar positions, and an exact search memory saves visits but almost no network evaluations.

1 Introduction

Engines in the AlphaGo line choose a move by a tree search guided by a policy and value network [Silver *et al.*, 2016; Silver *et al.*, 2017], and they are compared at a fixed number of search visits per move. KataGo [Wu, 2019], the strongest openly available engine of this kind, follows the convention: at 200 visits per move, a forced recapture and a whole-board fight receive the same 200 visits.

How much does that uniformity cost? Classical work on time management for Monte-Carlo tree search reports large gains from spending more time on some moves than on others

[Huang *et al.*, 2010; Baier and Winands, 2016]. For neural engines the evidence comes from small boards and small networks [Lan *et al.*, 2021; Ye *et al.*, 2022; Muppidi *et al.*, 2026], and KataGo already adapts its search inside the tree [KataGo project, 2026]. Whether a per-move budget decision still helps a modern, strong engine on the full board is an open empirical question.

We answer it with a wrapper, Mikiri, that changes nothing but the number of visits a move receives. It searches at 50 visits and asks a small model whether the decision is settled. If not, it searches at the next rung of a ladder, and asks again. The model predicts the winrate the current choice may still be giving up, and search stops when that stake, divided by the visits the next rung would add, falls below a threshold.

Our contributions are these.

- **A testbed with exact accounting.** Both players use the same engine, network and move-selection rule. Compute is reported as visits, as visits with every restarted search charged, and as network evaluations measured per player.
- **A ladder dataset with independent labels.** 6,000 positions, seven budgets each, every chosen move scored by a separate forced search. It shows that search error is rare and concentrated, and it forecasts match Elo.
- **A stopping rule that is worth nearly a doubling of search.** +228 Elo over KataGo at equal visits, and +93 Elo with 20% fewer.
- **A comparison with ten published rules** on one engine, offline and, for the strongest two, in play.
- **The rate rule.** Pricing the next rung matters more than the model, and improves prior signals.
- **Two negative results** on retrieval for a frozen engine.

2 Related Work

Deciding how long to search. Huang *et al.* [2010] extend search when the most-visited move is not the highest-valued one. Baier and Winands [2016] find that stopping early when the leader cannot be overtaken is the most reliable of several strategies, and Leela Chess Zero ships a variant of it together with a stop on the KL gain of the visit distribution [Leela Chess Zero contributors, 2026]. Hay *et al.* [2012] and Kaufmann and Koolen [2017] give stopping rules with a value-of-information and a best-arm-identification rationale. For

AlphaZero-style agents, DS-MCTS [Lan *et al.*, 2021] trains networks to predict whether the current choice will be worse than the final one, V-MCTS [Ye *et al.*, 2022] stops when virtually expanded policies have converged, and Muppidi *et al.* [2026] learn a budget chosen before search. These neural results are on 9×9 boards or small planners. We test all of them on 19×19 with a strong network (Section 6.1).

Reusing search. Transposition tables and graph search share statistics inside one search [Childs *et al.*, 2008; Czech *et al.*, 2021]; opening books do so across games [Wu, 2026]. Retrieval of similar positions helps agents whose networks are trained to read the neighbours [Humphreys *et al.*, 2022; Goyal *et al.*, 2022] and inside a single search [Xiao *et al.*, 2018]. We ask what a frozen engine can reuse.

3 Testbed

Engine and rules. Both players query KataGo v1.18.2 through its analysis engine. Each query is an independent search with an exact visit limit and one search thread, so no tree is inherited between moves and a visit means the same thing at every budget. The network is the 18-block b18c384nbt unless stated. Games use area scoring, positional superko and komi 7, the value at which the engine rates the empty board as even. Both players select moves as KataGo does by default, sampling in proportion to visits^{1/T} with T decaying from 0.5 to 0.1. A player resigns after three consecutive turns below 3% winrate.

Compute. Mikiri searches a position at 50 visits and then perhaps at 200. With one thread and cached evaluations the shallow tree is a prefix of the deep one: on fresh engines, a 200-visit search followed by a 400-visit search evaluated 366 to 396 network rows, against 353 to 395 for the 400-visit search alone. We nevertheless report three measures: *visits*, the size of the deepest search per move; *restart visits*, the sum of all searches requested; and *evaluations*, network rows executed by each player’s own engine process. Budgets are matched on visits.

Matches. A match is a series of games from the empty board with alternating colours. Intervals are bootstrap percentiles over colour-balanced pairs. KataGo against itself at 200 visits scores 50.2% over 400 games (+2 Elo).

4 The Ladder Dataset

We played 300 self-play games at budgets drawn log-uniformly from 30 to 200 visits and sampled 20 positions per game. Each position is searched at 50, 100, 200, 400, 800, 1,600 and 3,200 visits. A search cannot grade itself, so for every distinct move any rung chose we run a separate 1,600-visit search restricted to that move at the root. Its winrate $q(s, a)$ is our value of playing a . The *regret* of stopping at rung k is $r_k(s) = \max_a q(s, a) - q(s, a_k(s))$, where a_k is the move chosen at rung k and the maximum is over scored moves. For a policy with mean regret $\bar{r}$ we report the uniform budget with the same mean regret, interpolating the uniform curve in log visits. The ratio of that budget to the policy’s own mean cost is its *compute multiplier*.

Regret is rare and concentrated. At 200 visits the engine gives up 0.99% of winrate per move, and 71% of positions

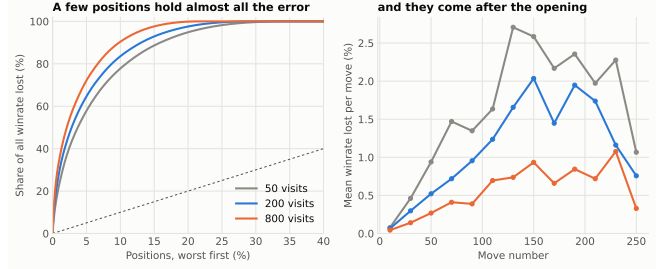


Figure 1: Left: share of all regret held by the worst positions. Right: mean regret by move number.

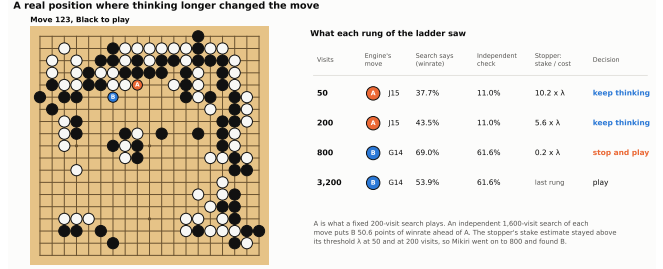


Figure 2: A position where the stopper keeps searching and the move changes. The threshold λ shown gives a mean cost of 200 visits.

carry no regret. The worst tenth hold 84% of it (Figure 1). Regret averages 0.19% per move over the first 40 moves and 1.4 to 1.6% after move 100. A stopper that knew the true regret would reach a multiplier of 12 at a 200-visit budget.

Figure 2 shows one of the costly positions. At 50 and at 200 visits the engine chooses a move that an independent search rates at 11%, and at 800 visits it finds one rated at 62%.

No single signal finds it. Table 1 asks how well individual search statistics rank positions by whether more than 0.5% of winrate remains on the table. Measures of how contested the root is reach an AUC of about 0.74. KataGo’s own prediction of its short-term winrate error, which it uses inside the tree to weight playouts, reaches 0.58 to 0.60. The classical cue that the most-visited move is not the highest-valued one [Huang *et al.*, 2010] reaches 0.59 to 0.61. Contested positions are common, and most are contested between moves of nearly equal value, where being wrong costs nothing.

5 Mikiri

The rate rule. Let the ladder have rungs $v_1 < \dots < v_m$, and let a stopping policy π stop position s at rung $\pi(s)$. We want the policy that minimises expected regret subject to an expected cost,

$$\min_{\pi} \mathbb{E}[r_{\pi(s)}(s)] \quad \text{s.t.} \quad \mathbb{E}[v_{\pi(s)}] \leq B. \quad (1)$$

With a multiplier λ on the constraint, the problem separates over positions: each position minimises $r_k(s) + \lambda v_k$. At rung j the visits already spent cannot be recovered, and continuing by one rung is worthwhile exactly when $\mathbb{E}[r_j - r_{j+1} \mid x] > \lambda(v_{j+1} - v_j)$, where x is what the finished search shows. The reduction $r_j - r_{j+1}$ is at most r_j . Replacing it by a prediction

How Mikiri decides one move

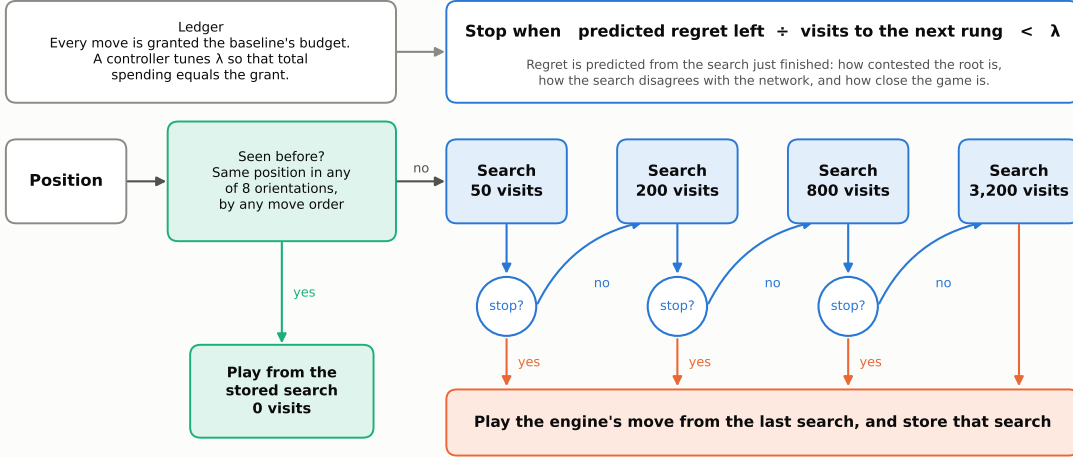


Figure 3: How Mikiri decides one move.

Signal	50	200	800
Top move's visit share	0.75	0.74	0.74
LCB margin	0.74	0.73	0.73
v1 entropy gate	0.73	0.73	0.74
KL(visits, prior)	0.65	0.63	0.61
Most-visited is not best-valued	0.60	0.59	0.61
KataGo's predicted winrate error	0.58	0.58	0.60
Searched minus raw winrate	0.57	0.56	0.56

Table 1: AUC for ranking positions by remaining regret above 0.5%, from the search at 50, 200 and 800 visits.

167 $\hat{r}_j(x)$ of the regret remaining gives the one-step rule: stop
168 when

$$\frac{\hat{r}_j(x)}{v_{j+1} - v_j} < \lambda. \quad (2)$$

169 One λ prices every decision on the ladder alike. On a geomet-
170 ric ladder the next rung costs more each time, so going from
171 800 to 3,200 visits needs four times the stake that going from
172 200 to 800 does. A flat threshold on $\hat{r}_j$ ignores that price, and
173 spends most of a long ladder's budget on a few positions that
174 no amount of search resolves.

175 **The model.** $\hat{r}$ is a gradient-boosted tree ensemble (150
176 trees of depth 3) fitted to the square root of regret, with all non-
177 final rungs pooled. Its 30 inputs come from the finished search
178 at no engine cost: how contested the root is (visit shares, ent-
179ropy, the margin between the best move's lower confidence
180 bound and its best rival), how the search disagrees with the
181 network (priors, KL from prior to visits, raw against searched
182 winrate), what is at stake (winrate, distance from 50%, score),
183 KataGo's own raw error predictions, game phase, and what
184 changed since the previous rung. All offline numbers are
185 cross-fitted by game.

186 **The ledger.** Mikiri is granted the baseline's budget B for
187 every move, and a ledger records what it spends. The off-
188 line frontier gives a map $\lambda(c)$ from a target mean cost c to

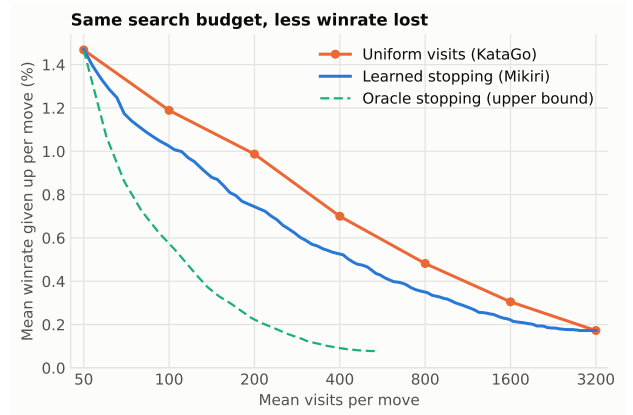


Figure 4: Mean regret against mean visits on the four-rung ladder.

189 a threshold. After each move the controller sets the next tar-
190 get to B plus the ledger's balance spread over a horizon of
191 3,000 moves, and divides it by a running ratio of realised to
192 commanded spending, which corrects for the difference be-
193 tween the dataset and real games. Spending tracks the grant
194 to within 1% in every match we report.

195 **Memory.** Finished searches from the first 80 moves are
196 stored under a key that is invariant to the eight board symme-
197 tries and to move order. A later hit is played from the stored
198 search at zero visits.

6 Offline Results

199 **Multiplier.** On the four-rung ladder (50, 200, 800, 3,200)
200 the stopper at a mean cost of 200 visits loses as little as uni-
201 form search at 358, a multiplier of 1.80, and stays between
202 1.69 and 1.88 from 100 to 800 visits (Figure 4). With every
203 restart charged it is 1.29 to 1.51. On the full doubling ladder
204 it reaches 2.13.
205

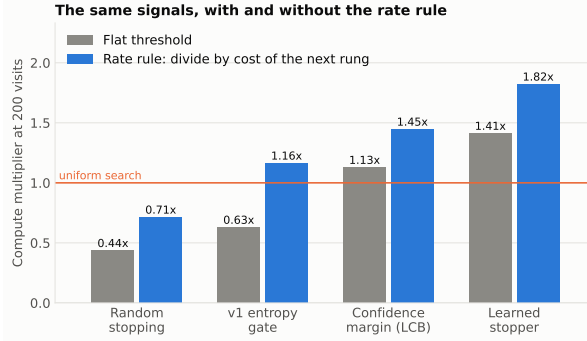


Figure 5: Four stopping signals on the four-rung ladder at 200 visits, with a flat threshold and with the rate rule.

What matters. Removing the trajectory features, KataGo’s raw error predictions, or all but seven features changes the multiplier at 200 visits by at most 0.09, and 300 labeled positions already give 1.50. Fitting the square root of regret instead of regret raises it from 1.74 to 1.82. Replacing the rate rule by a flat threshold lowers it to 1.41. Figure 5 shows the same contrast for hand-written signals: with a flat threshold the entropy gate of an earlier prototype is worse than no gate (0.63), and with the rate rule it is a gain (1.16).

6.1 Comparison with Prior Rules

We re-implemented ten published rules from their papers and source code (Table 2). All see the same root statistics and are scored by the same forced values. *BEHIND* and *UNST* search longer when the player is losing or when the most-visited move is not the highest-valued one [Huang *et al.*, 2010]. *STOP* ends a search once the runner-up cannot overtake the leader in the visits that remain, or in a fraction p of them, and *CLOSE* extends it when the top two moves are close in visits [Baier and Winands, 2016]. *Smart pruning* is Leela Chess Zero’s form of *STOP*, shipped with a stop on the per-visit KL gain of the visit distribution [Leela Chess Zero contributors, 2026]. The *VOI* rule stops when a bound on the value of further samples falls below their cost [Hay *et al.*, 2012], and the *BAI* rule when the leader’s confidence interval separates from every rival’s [Kaufmann and Koolen, 2017]. The *state-only* rule predicts regret from features available before any search, in the spirit of Muppidi *et al.* [2026]. Two learned rules are adaptations: for DS-MCTS we train a classifier on root statistics to the authors’ label $U(s, n)$, in place of their board-input networks, and for V-MCTS we run the authors’ virtual expansion on KataGo’s root priors, visits and winrates with their published $r = 0.2$ and $\epsilon = 0.1$. Every rule is charged continuation cost, which is what a rule inside the engine pays. Charged for every restart, Mikiri’s four-rung ladder reaches 1.44 at 200 visits, below the 1.60 of V-MCTS, which has no restarts to charge; the matches settle that question in network evaluations.

Three findings. First, the rules with a statistical stopping guarantee are far worse than uniform search here, because they stop on confidence in the leading move and ignore whether the game is still in doubt. Second, V-MCTS is the strongest prior rule (1.60 at 200 visits), and Mikiri exceeds it by 0.54 [0.28, 0.84] at 200 visits and 0.35 [0.09, 0.65] at 400, paired over the

same resampled games. Third, the rate rule improves every signal it can be applied to, including the DS-MCTS classifier (1.48 to 1.75), which with it ties Mikiri at 400 visits (0.03 [-0.16, 0.27]).

What to predict, and how to price it. Remaining regret is an upper bound on what continuing can recover. Table 3 compares it with the quantities the derivation names: the gain from the next rung, the gain from the whole ladder, and the best recovery per visit over any continuation, $\max_{k>j}(r_j - r_k)/(v_k - v_j)$. Read by an oracle, that last index reproduces the hindsight optimum (14.11 against 14.16). Learned from search statistics it trails remaining regret under the rate rule (1.63 to 1.87 against 1.83 to 2.14). The next-rung gain is weakest: myopic as an oracle, noisy as a target. Pricing remaining regret by the visits to the top rung is worse than no price. The rate rule pairs a target that search statistics can predict with the one cost that is known exactly.

Do prior rules see more? We fed the KLD gain, the VOI bound, the BAI gap and the V-MCTS distance to Mikiri’s regressor as extra inputs, singly and together. The multiplier at 200 visits moved from 2.14 to between 2.07 and 2.17, with no paired gain distinguishable from zero. The rules differ in how they use the root statistics; the statistics themselves are exhausted.

7 Matches

Equal visits. With the six-rung ladder (50, 200, 400, 800, 1,600, 3,200) Mikiri scores 78.8% (750–175–75), a gain of +228 Elo (+203 to +254). With the four-rung ladder it gains +206 (+181 to +232) over 1,000 games. KataGo at 400 visits gains +237 (Table 4). Played directly against KataGo at 400 visits, Mikiri at 200 is level: 268–271–61 over 600 games, –2 Elo (–27 to +24), with 102 evaluations per move to KataGo’s 143.

Fewer visits and equal evaluations. A fixed-budget engine finds much of its next tree already cached, and Mikiri’s varying searches reuse less, so it runs more evaluations per visit. Granted 160 visits, it requests fewer visits than KataGo even with restarts charged (197 against 200) and gains +93 Elo (+72 to +114). At close to equal evaluations the comparison is direct: Mikiri runs 102 per move and gains +228, KataGo at 283 visits runs 98 and gains +108 (Figure 6).

No move sampling. A search that stops when one move dominates has sharper visit counts, which could interact with sampling. With 400 balanced openings each played twice with colours swapped, and both players always playing the top move, Mikiri gains +178 Elo (+153 to +205).

The forecast. Reading the cross-fitted regret at the cost Mikiri actually used, converting to equivalent uniform visits, and reading that budget’s Elo off the measured uniform curve predicts +197 for the four-rung match against the measured +206, +126 at 160 visits against +93, +225 for the six-rung ladder against +228, +145 for the V-MCTS rule against +135, and +128 for the DS-MCTS-style rule against +146. The forecast is within 35 Elo of each and places both prior rules below Mikiri’s equal-visit matches. Its one clear miss is Mikiri’s rule on the short V-MCTS ladder, forecast at +196 and measured at +150 (Table 5). The ladder dataset is a useful

Rule	As published, 200	As published, 400	With rate rule, 200
VOI stop [Hay <i>et al.</i> , 2012]	0.34 [0.30, 0.40]	0.24 [0.20, 0.31]	1.05 [0.86, 1.21]
BAI stop [Kaufmann and Koolen, 2017]	0.77 [0.63, 1.01]	0.59 [0.50, 0.71]	1.18 [1.09, 1.29]
BEHIND [Huang <i>et al.</i> , 2010]	0.89 at 174	0.89 at 348	–
STOP, $p = 0.45$ [Baier and Winands, 2016]	1.25 at 134	1.19 at 512	–
Smart pruning [Leela Chess Zero contributors, 2026]	1.17 at 159	1.21 at 308	–
CLOSE [Baier and Winands, 2016]	1.15 at 229	1.11 at 453	–
UNST [Huang <i>et al.</i> , 2010]	1.21 at 239	1.17 at 475	–
KLD gain [Leela Chess Zero contributors, 2026]	1.32 [1.18, 1.49]	1.23 [1.02, 1.42]	–
State only [Muppidi <i>et al.</i> , 2026]	1.18 [0.97, 1.40]	1.02 [0.89, 1.23]	1.54 [1.35, 1.74]
DS-MCTS [Lan <i>et al.</i> , 2021]	1.48 [1.28, 1.73]	1.48 [1.29, 1.73]	1.75 [1.60, 1.92]
V-MCTS [Ye <i>et al.</i> , 2022]	1.60 [1.44, 1.76]	1.54 [1.36, 1.70]	–
LCB margin	1.14 [0.95, 1.36]	0.82 [0.70, 0.97]	1.45 [1.24, 1.70]
Mikiri	1.40 [1.21, 1.63]	1.35 [1.15, 1.61]	2.13 [1.88, 2.41]

Table 2: Compute multipliers on the doubling ladder (above 1 is better than uniform search), with 95% bootstrap intervals over games. Threshold rules are swept and shown at their best point for each budget; rules without a threshold are shown at the configuration nearest the budget, with its cost in visits.

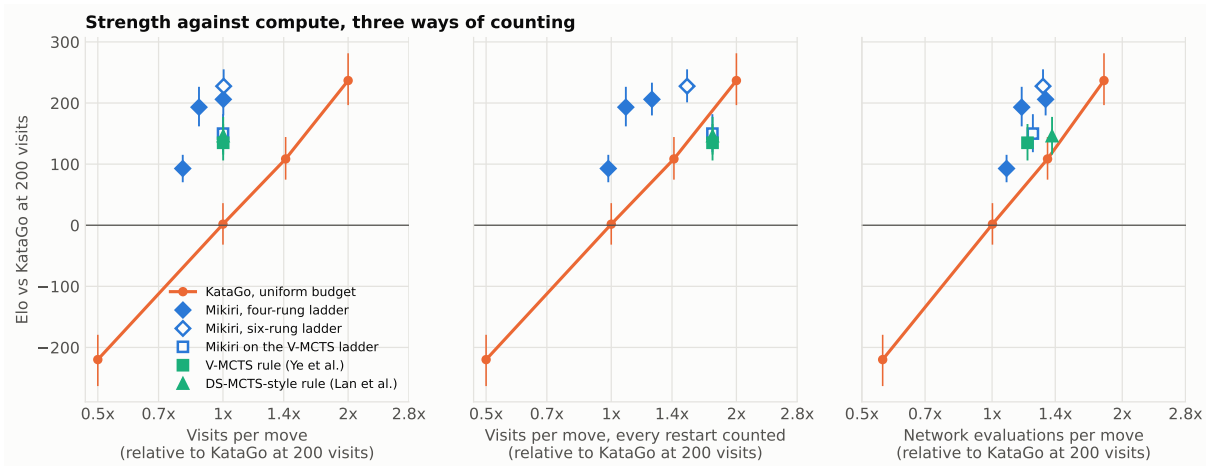


Figure 6: Elo against KataGo at 200 visits as a function of compute relative to that baseline, counted three ways. The line is KataGo at uniform budgets; bars are 95% intervals.

guide for choosing among rules and no substitute for playing the games.

Prior rules in play. The V-MCTS rule gains +135 Elo (+108 to +164) and the DS-MCTS-style classifier +146 (+117 to +176), each held to 200 visits by the same controller. The DS-MCTS-style rule runs 105 evaluations per move, more than Mikiri’s six-rung ladder, so Mikiri is ahead of it on every measure. At equal visits Mikiri is also ahead of V-MCTS, with intervals that do not overlap. In evaluations the gap closes: V-MCTS, whose checkpoints stop at 400 visits, runs 87 per move to Mikiri’s 99, and interpolating Mikiri’s 160- and 200-visit matches to 87 evaluations gives about +135, level with V-MCTS. With Mikiri’s stopper on V-MCTS’s own ladder (50, 100, 200, 400), so that the rules differ in nothing else, it gains +150 (+121 to +180) at 92 evaluations, which the match cannot separate from V-MCTS. Mikiri’s lead at equal visits comes from ladders that reach 3,200 visits, which the rate rule makes affordable and which cost more evaluations per visit. At matched evaluations, with the four-rung ladder granted 175 visits, Mikiri gains +193

(+163 to +225) at 89 evaluations to V-MCTS’s +135 at 87: a lead of about 60 Elo whose intervals touch. The LCB margin with the rate rule and no learning gains +88 (+61 to +115), and the entropy gate with a flat threshold ?? (?? to ??).

Where the visits go. Mikiri uses a third of the budget over the first 20 moves and about 270 visits per move around move 140. After move 60 it spends about 130 visits when it rates its winrate above 90% and up to 330 when it is behind but not lost (Figure 7). It scores 76.2% as Black and 77.0% as White. Neither behaviour was programmed; both follow from predicting regret in winrate units.

Other settings. The same model, without retraining, was played in further settings (Table 6).

8 What a Frozen Engine Can Retrieve

Exact memory. Positions recur: across 300 self-play games, a position at move 20 had occurred in an earlier game, up to symmetry, half of the time. In play, memory alone served 21.2% of moves over 600 games with strength unchanged (−9 Elo, −37 to +19) at 158 visits per move. It did not reduce

Scored	Divided by	Oracle, 100 visits	Learned, 200 visits			
			4	6	7	rungs
r_j	1 (flat)	5.99	1.41	1.56	1.51	
r_j (rate rule)	$v_{j+1} - v_j$	8.69	1.83	1.98	2.14	
r_j	$v_m - v_j$	5.86	1.35	1.50	1.39	
$r_j - r_{j+1}$	$v_{j+1} - v_j$	4.96	1.50	1.54	1.44	
$r_j - r_m$	$v_{j+1} - v_j$	9.07	1.79	1.98	2.04	
$\max_{k>j} \frac{r_j - r_k}{v_k - v_j}$	built in	14.11	1.63	1.87	1.77	
Hindsight optimum		14.16	—	—	—	

Table 3: Multipliers for different quantities to score at rung j and prices to divide them by (m is the top rung). Oracle: true quantity, four-rung ladder, 100 visits. Learned: same model fitted to each quantity, 200 visits, three ladders.

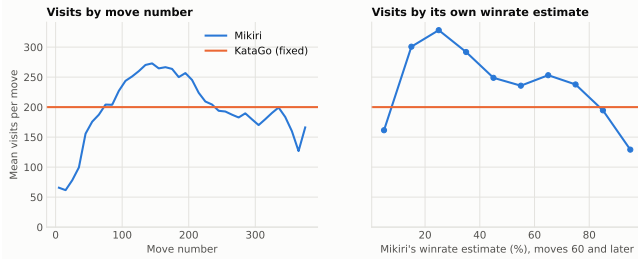


Figure 7: Mean visits per move in the four-rung match, by move number and by Mikiri’s own winrate estimate.

network evaluations (69 per move against 69): KataGo caches evaluations for the life of its process, so the positions memory serves were already cheap.

Approximate retrieval. We embed positions with the raw policy and ownership maps, align over the eight symmetries, and offer the nearest stored position’s 3,200-visit move as a proposal. Where the 200-visit and 3,200-visit moves differ (1,605 positions), the nearest neighbour’s move is right 5.5% of the time, against 43% for the policy network’s own best alternative. A neighbour’s move matches in 96% of cases at similarity above 0.999, which is the same position, and in 27% at 0.90 to 0.97. Neighbours’ regret predicts a position’s regret with AUC 0.55.

9 Limitations

Results are for KataGo networks at 100 to 800 visits, against KataGo, where a doubling of visits is worth about 230 Elo; larger budgets are untested. Regret is defined by a search and inherits its blind spots, though the matches do not depend on it. Of the regret that survives a 200-visit search, 57% sits on moves that search gave under 10% of its visits, which bounds what any rule reading root statistics can recover. Our versions of prior rules are re-implementations on a different engine than their authors used. A stopping rule built on the engine’s own statistics offers no protection against adversarial policies [Wang *et al.*, 2023].

10 Conclusion

A frozen Go engine leaves most of a doubling of search on the table by spending visits uniformly. A small learned stopping rule recovers it, and the part of the rule that matters most is the price it puts on continuing.

Ethical Statement

There are no ethical issues.

Use of AI Assistance

A large language model coding assistant was used extensively in this work: to implement the testbed and analysis code, to run and monitor the experiments, to produce the figures, and to draft and edit the text. The authors directed the work and are responsible for all of its content. Every reported number is produced by released scripts from released data and game records.

References

- [Baier and Winands, 2016] Hendrik Baier and Mark H. M. Winands. Time management for Monte Carlo tree search. *IEEE Transactions on Computational Intelligence and AI in Games*, 8(3):301–314, 2016.
- [Childs *et al.*, 2008] Benjamin E. Childs, James H. Brodeur, and Levente Kocsis. Transpositions and move groups in Monte Carlo tree search. In *IEEE Symposium on Computational Intelligence and Games*, pages 389–395, 2008.
- [Czech *et al.*, 2021] Johannes Czech, Patrick Korus, and Kristian Kersting. Improving AlphaZero using Monte-Carlo graph search. In *International Conference on Automated Planning and Scheduling*, pages 103–111, 2021. arXiv:2012.11045.
- [Goyal *et al.*, 2022] Anirudh Goyal, Abram L. Friesen, Andrea Banino, Theophane Weber, Nan Rosemary Ke, Adrià Puigdomènech Badia, Arthur Guez, Mehdi Mirza, Peter C. Humphreys, Ksenia Konyushkova, Laurent Sifre, Michal Valko, Simon Osindero, Timothy Lillicrap, Nicolas Heess, and Charles Blundell. Retrieval-augmented reinforcement learning. In *International Conference on Machine Learning*, pages 7740–7765, 2022. arXiv:2202.08417.
- [Hay *et al.*, 2012] Nicholas Hay, Stuart Russell, David Tolpin, and Solomon Eyal Shimony. Selecting computations: Theory and applications. In *Uncertainty in Artificial Intelligence*, 2012. arXiv:1207.5879.
- [Huang *et al.*, 2010] Shih-Chieh Huang, Rémi Coulom, and Shun-Shii Lin. Time management for Monte-Carlo tree search applied to the game of Go. In *International Conference on Technologies and Applications of Artificial Intelligence*, pages 462–466, 2010.
- [Humphreys *et al.*, 2022] Peter C. Humphreys, Arthur Guez, Olivier Tieleman, Laurent Sifre, Théophane Weber, and Timothy Lillicrap. Large-scale retrieval for reinforcement learning. In *Advances in Neural Information Processing Systems*, 2022. arXiv:2206.05314.

Player (vs. KataGo at 200 visits)	Games	W-L-D	Elo (95% interval)	Visits	Restart	Evals
KataGo, 100 visits	400	77-301-22	-220 (-262, -181)	100	100	40 : 71
KataGo, 283 visits	400	247-126-27	+108 (+76, +143)	283	283	98 : 73
KataGo, 400 visits	400	310-73-17	+237 (+198, +280)	400	400	132 : 73
Mikiri, seven-rung ladder			(running)			
Mikiri, six-rung ladder	1,000	750-175-75	+228 (+203, +254)	201	304	102 : 78
Mikiri, four-rung ladder	1,000	728-196-76	+206 (+181, +232)	200	251	99 : 74
Mikiri, four-rung, 175-visit grant	600	432-129-39	+193 (+163, +225)	175	217	89 : 76
Mikiri, four-rung, 160-visit grant	1,000	586-325-89	+93 (+72, +114)	160	197	80 : 74
Mikiri, four-rung, 140-visit grant			(running)			
Mikiri, four-rung, no move sampling	800	561-183-56	+178 (+153, +205)	200	250	115 : 96
V-MCTS rule (Ye et al.)	600	395-173-32	+135 (+108, +164)	200	350	87 : 72
Mikiri's rule on the V-MCTS ladder	600	404-160-36	+150 (+121, +180)	200	350	92 : 74
DS-MCTS-style rule (Lan et al.)	600	395-157-48	+146 (+117, +176)	200	351	105 : 77
LCB margin with rate rule, no learning	600	349-201-50	+88 (+61, +115)	200	250	100 : 75
Entropy gate, flat threshold			(running)			
Memory only	600	276-292-32	-9 (-37, +19)	158	158	69 : 69
Mikiri four-rung + memory	1,200	908-219-73	+227 (+204, +250)	201	255	109 : 76

Table 4: Matches against KataGo at 200 visits per move, 19×19 , same network. *Visits* is the mean size of the deepest search per move, *Restart* charges every search requested, *Evals* is network evaluations per move for the player and its opponent.

Rule	Visits	Equiv.	Forecast	Measured (95%)
Mikiri, 4 rungs	160	297	+126	+93 (+72, +114)
Mikiri, 4 rungs	175	325	+160	+193 (+163, +225)
Mikiri, 4 rungs	200	360	+197	+206 (+181, +232)
Mikiri, 6 rungs	201	387	+225	+228 (+203, +254)
Mikiri, short ladder	200	358	+196	+150 (+121, +180)
V-MCTS	200	312	+145	+135 (+108, +164)
DS-MCTS-style	200	299	+128	+146 (+117, +176)

Table 5: Elo forecast from the ladder dataset against Elo measured in play, versus KataGo at 200 visits.

Opponent visits	Games	W-L-D	Elo (95% interval)
100	400	262-119-19	+130 (+95, +167)
400		(running)	
800		(running)	
400 (Mikiri at 200)	600	268-271-61	-2 (-27, +24)
200, b28 network		(running)	
200, 13×13		(running)	
200, 9×9		(running)	

Table 6: The four-rung stopper, trained once on 19×19 b18 positions, against KataGo at matched visits in other settings.

namic simulation Monte-Carlo tree search. In *AAAI Conference on Artificial Intelligence*, 2021. arXiv:2012.07910. 431 432

[Leela Chess Zero contributors, 2026] Leela Chess Zero contributors. Lc0 options. <https://github.com/LeelaChessZero/lc0/wiki/Lc0-options>, 2026. Accessed 2026-10-01. 433 434 435 436

[Muppidi et al., 2026] Aneesh Muppidi, Firas Darwish, Dylan Cope, João F. Henriques, and Jakob Nicolaus Foerster. Finding the time to think: Learning planning budgets in real-time RL. *arXiv preprint arXiv:2606.26463*, 2026. 437 438 439 440

[Silver et al., 2016] David Silver, Aja Huang, Chris J. Maddison, Arthur Guez, Laurent Sifre, George van den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, Sander Dieleman, Dominik Grewe, John Nham, Nal Kalchbrenner, Ilya Sutskever, Timothy Lillicrap, Madeleine Leach, Koray Kavukcuoglu, Thore Graepel, and Demis Hassabis. Mastering the game of Go with deep neural networks and tree search. *Nature*, 529:484-489, 2016. 441 442 443 444 445 446 447 448 449

[Silver et al., 2017] David Silver, Julian Schrittwieser, Karen Simonyan, Ioannis Antonoglou, Aja Huang, Arthur Guez, Thomas Hubert, Lucas Baker, Matthew Lai, Adrian Bolton, Yutian Chen, Timothy Lillicrap, Fan Hui, Laurent Sifre, George van den Driessche, Thore Graepel, and Demis Hassabis. Mastering the game of Go without human knowledge. *Nature*, 550:354-359, 2017. 450 451 452 453 454 455 456

[Wang et al., 2023] Tony T. Wang, Adam Gleave, Tom Tseng, Kellin Pelrine, Nora Belrose, Joseph Miller, Michael D. Dennis, Yawen Duan, Viktor Pogrebniak, Sergey Levine, and Stuart Russell. Adversarial policies beat superhuman Go AIs. In *International Conference on Machine Learning*, pages 35655-35739, 2023. arXiv:2211.00241. 457 458 459 460 461 462 463

[KataGo project, 2026] KataGo project. Other methods implemented in KataGo. <https://github.com/lightvector/KataGo/blob/master/docs/KataGoMethods.md>, 2026. Accessed 2026-10-01. 421 422 423 424

[Kaufmann and Koolen, 2017] Emilie Kaufmann and Wouter M. Koolen. Monte-Carlo tree search by best arm identification. In *Advances in Neural Information Processing Systems*, 2017. arXiv:1706.02986. 425 426 427 428

[Lan et al., 2021] Li-Cheng Lan, Meng-Yu Tsai, Ti-Rong Wu, I-Chen Wu, and Cho-Jui Hsieh. Learning to stop: Dy- 429 430

- 464 [Wu, 2019] David J. Wu. Accelerating self-play learning in
465 Go. *arXiv preprint arXiv:1902.10565*, 2019.
- 466 [Wu, 2026] David J. Wu. KataGo opening books. [https://](https://katagobooks.org/)
467 katagobooks.org/, 2026. Accessed 2026-10-01.
- 468 [Xiao *et al.*, 2018] Chenjun Xiao, Jincheng Mei, and Martin
469 Müller. Memory-augmented Monte Carlo tree search. In
470 *AAAI Conference on Artificial Intelligence*, 2018.
- 471 [Ye *et al.*, 2022] Weirui Ye, Pieter Abbeel, and Yang Gao.
472 Spending thinking time wisely: Accelerating MCTS with
473 virtual expansions. In *Advances in Neural Information Pro-*
474 *cessing Systems*, 2022. arXiv:2210.12628.